Solar Al appendix

Sun · Al · downloadable appendix

Document	Solar Al appendix
Subject	Solar photospheric abundance appendix · Al
Author	Ryan Schmitt
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Web	https://exoplanetcodex.org/systems/sol/elements/al/

Solar aluminium: 6.429 ± 0.081

Asplund, Amarsi & Grevesse 2021: 6.43 ± 0.03 · this measurement -0.001 dex

HARPS reaches only two of these lines, and its continuum placement there is less certain than on the Kitt Peak atlas, so this is our Kitt Peak product on the same lines with the same treatment. Products and engines are not averaged together.

VIS: 6.429 ± 0.081 · Kitt Peak (Kurucz 2005) · VIS · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Highlighted Products

VIS

6.429 ± 0.081

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

red-optical

6.450 ± 0.083

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

6.552 ± 0.144

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

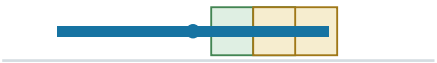
Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	16 products -- Al I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	14 products -- Al I · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

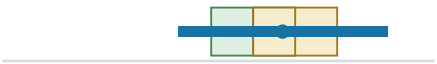
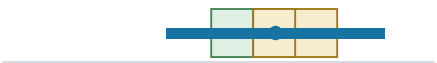
VIS — HARPS
A(Al) dex; axis 6.25–6.56

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.409 ±0.093
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.387 ±0.097
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.403 ±0.096
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.381 ±0.100

VIS — IAG FTS
A(Al) dex; axis 6.25–6.56

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.447 ±0.082
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.425 ±0.088
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.443 ±0.087
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.420 ±0.092

VIS — Kitt Peak (Kurucz 2005)
A(Al) dex; axis 6.25–6.56

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.451 ±0.075
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.429 ±0.081
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.446 ±0.078
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.423 ±0.084

VIS — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.25–6.56

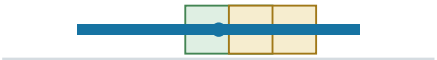
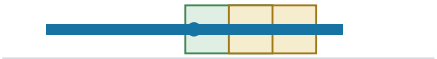
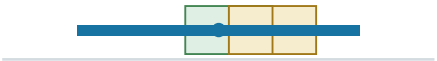
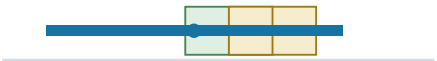
Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.450 ±0.075
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.428 ±0.081

VIS — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.25–6.56

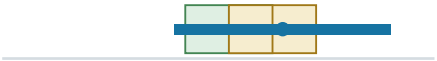
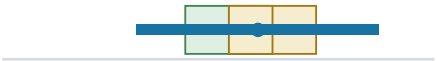
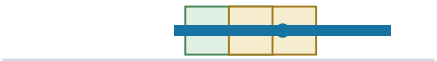
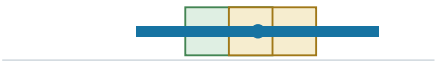
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.450 ±0.073
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.427 ±0.079

VIS axis: 6.25 · 6.33 · 6.41 · 6.48 · 6.56 A(Al) dex

red-optical — IAG FTS
A(Al) dex; axis 6.27–6.57

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.423 ±0.097
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.406 ±0.102
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.423 ±0.097
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.406 ±0.102

red-optical — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.27–6.57

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.467 ±0.074
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.450 ±0.083
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.467 ±0.074
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.450 ±0.083

red-optical axis: 6.27 · 6.35 · 6.42 · 6.50 · 6.57 A(Al) dex

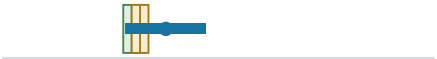
NIR — CRIRES+ Y (full arm)
A(Al) dex; axis 5.97–7.51

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		6.747 ±0.733
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		6.732 ±0.734

NIR — Kitt Peak (1984, molecfit)
A(Al) dex; axis 5.97–7.51

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.569 ±0.141

NIR — Kitt Peak (1984, molecfit)
A(Al) dex; axis 5.97–7.51

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.552 ±0.144
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.569 ±0.141
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.552 ±0.144

NIR axis: 5.97 · 6.35 · 6.74 · 7.12 · 7.51 A(Al) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the total uncertainty -- the statistical and systematic terms combined in quadrature, as Asplund et al. 2021 and Lodders et al. 2025 quote it. The full error budget remains attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every aluminium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the aluminium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Al)	Largest uncertainty terms (dex)
VIS	Al I · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	6.381–6.427 (1D-NLTE)	line strengths (gf) 0.06, continuum placement 0.04
VIS	Al I · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	6.387–6.429 (1D-NLTE)	line strengths (gf) 0.06, continuum placement 0.04
red-optical	Al I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	2	6.406–6.450 (1D-NLTE)	fit / line-to-line scatter 0.06, line strengths (gf) 0.05
red-optical	Al I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	2	6.406–6.450 (1D-NLTE)	fit / line-to-line scatter 0.06, line strengths (gf) 0.05
NIR	Al I · Codex-graded lines	Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	2	6.552 (1D-NLTE)	fit window 0.11, continuum placement 0.06

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Al)	Largest uncertainty terms (dex)
NIR	Al I · Asplund et al. 2021 lines	CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–2	6.552–6.732 (1D-NLTE)	fit window 0.11, continuum placement 0.06

Solar aluminium context

Aluminium is traced by Al I lines in the optical and near-infrared. The reference lines are the six Asplund et al. 2021 used (Nordlander & Lind 2017: 6696, 6698, 7835, 8912, 10768 and 10872 Å) with the oscillator strengths and uncertainties those authors adopted. Al I departs from LTE, so each spectrum also carries a non-LTE-corrected product; each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Al appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 27afcfcba7db6b4782ca756c1131fb9ddaa8dc3c
Feed Al.json v1.67 · updated 2026-10-16T12:38:38Z
Generated 2026-10-28T11:24:25-06:00 (source commit timestamp)
Feed SHA-256 09b3bed9e91ec7f33ab4903c5bd78089abe4f87946052bba1bff91c864638978
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Al comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set AL_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Al I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Al_Amarsi2020_PySME.csv). Sources: [5].

Product-to-source index

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-3dced287aa93

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-6a4b66082042

NIR · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-e1f71442de77

NIR · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-9443d9b76f0f

NIR · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-2c817b8eea2e

NIR · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-fcec422376c8

VIS · HARPS · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-49c83f2985f1

VIS · HARPS · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-cfa1cc6e0651

VIS · HARPS · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-8d6674191138

VIS · HARPS · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-f02ef045fac9

VIS · IAG FTS · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-0f1b0ce78149

VIS · IAG FTS · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-b53b771ad05b

VIS · IAG FTS · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-5413d8634cb7

VIS · IAG FTS · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-eaef832d7490

VIS · Kitt Peak (Kurucz 2005) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-b526417c1fab

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VIS · Kitt Peak (Kurucz 2005) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-fd94f2a34bc8

VIS · Kitt Peak (Kurucz 2005) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-0ecd4c989e62

VIS · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-f1ad198137d1

VIS · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-e549e5560416

VIS · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-ed888010e686

VIS · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-4113208fe244

red-optical · IAG FTS · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-5de0eeb4504d

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red-optical · IAG FTS · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-a86d8aaf060b

red-optical · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-b130e20bd3cc

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red-optical · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-512ac5762764

red-optical · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-1636173039ab

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

[3] — Nordlander, T.; Lind, K. (2017). Non-LTE aluminium abundances in late-type stars. A&A 607, A75 DOI: 10.1051/0004-6361/201730427. source record

Model and grid sources

[4] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[5] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record